

Draft Proposal

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Proposal 1 – Generative Networks for Shape Completion (22)

The idea of the project is to train a fast and lightweight latent diffusion model that would be used for 3D shape completion. The model would consist of a variational autoencoder and a diffusion transformer model.

The main inspiration for the variational autoencoder stems from [1]. We would implement the autoencoder as specified in the paper with the possibility of changing some parts of the architecture. The hybrid triplane representation would help us distill the 3D information to a smaller space of high-resolution images and a coarse 3D occupancy map.

For the diffusion transformer we would implement a lightweight model. The architectures mentioned in [2], [3], and [4] could be used as guidance for the model architecture, but we would also experiment with different architectures.

The training would be done on the ShapeNet dataset, starting the training only on one object category and later expanding to a reasonable number given the training resources. The 3D model of the objects would be randomly occluded either before the training process or on the fly, depending on the memory size of the dataset. The objects can be occluded in such a way as to mimic a real-world point cloud from an active camera or a lidar sensor. Also, there is a possibility of adding noise to our training dataset in order to make our model more robust for a real-world scenario.

Since we chose a diffusion transformer, we could implement an uncertainty score of the correct shape reconstruction.

References:

- [1] Jingyu Guo, Sensen Gao, Jia-Wang Bian, Wanhu Sun, Heliang Zheng, Rongfei Jia, Mingming Gong. Hyper3D: Efficient 3D Representation via Hybrid Triplane and Octree Feature for Enhanced 3D Shape Variational Auto-Encoders. arXiv preprint arXiv:2503.10403, 2025
- [2] Shentong Mo, Enze Xie, Ruihang Chu, Lewei Yao, Lanqing Hong, Matthias Nießner, Zhenguo Li. DiT-3D: Exploring Plain Diffusion Transformers for 3D Shape Generation. arXiv preprint arXiv:2307.01831, 2023
- [3] Shuang Wu, Youtian Lin, Feihu Zhang, Yifei Zeng, Jingxi Xu, Philip Torr, Xun Cao, Yao Yao. Direct3D: Scalable Image-to-3D Generation via 3D Latent Diffusion Transformer. arXiv preprint arXiv:2405.14832, 2024
- [4] William Peebles, Saining Xie. Scalable Diffusion Models with Transformers. arXiv preprint arXiv:2212.09748, 2023

Proposal 2 – Generative Networks for Robot Grasping (5)

Grasp synthesis is one of the most important problems in robotic manipulation. In this task a robot observes an object and needs to predict a gripper pose that leads to a stable grasp. Classical analytical methods rely on precise geometrical models and contact physics, but they generalize poorly on new unknown scenes and real world noise [1]. Recent work proposes data driven generative strategies, learning valid 6-DoF grasp poses conditioned on sensory input such as point clouds [2].

An early approach by 6-DoF GraspNet uses a variational autoencoder (VAE) [1]. While achieving very promising results, the encoder generates many invalid grasps and therefore the generator required an iterative refinement process via an additional evaluator network [1]. More recent methods based on diffusion models significantly improve grasp quality and diversity.

In particular, GraspLDM: Generative 6-DoF Grasp Synthesis Using Latent Diffusion Models performs diffusion in the latent space of a VAE, enabling efficient training and high-quality grasp generation. The framework supports conditional generation, allowing the inclusion of task-specific or geometrical constraints [2]. Interesting research areas for GraspLDM could be to incorporate region constraints into the grasp synthesis that are not coupled to any tasks. Lundell *et al.* modified the ACRONYM [4]

Interesting research directions could explore incorporating region-based constraints to the grasp synthesis procedure as investigated by Lundell *et al.* as well as investigating alternative diffusion formulations such as Denoising Diffusion Implicit Model (DDIM) to improve efficiency.

- Preprocess the ACRONYM: A large-scale simulation-based grasp dataset
- Implement a VAE based on GraspNet as a baseline and evaluate generated grasps
- Extend the model to Latent Diffusion architecture (GraspLDM-style)
- Compare VAE and Latent Diffusion in terms of success rate, diversity and inference time
- Introduce region-based constraints for conditional grasp generation
- Implement implicit diffusion as an alternative to standard diffusion and investigate its impact on performance

References:

- [1] A. Mousavian, C. Eppner, and D. Fox, "6-DOF GraspNet: Variational Grasp Generation for Object Manipulation," 2019. [Online]. Available: <https://arxiv.org/pdf/1905.10520>
- [2] K. Barad, A. Orsula, and A. Richard, "GraspLDM: Generative 6-DoF Grasp Synthesis Using Latent Diffusion Models,"
- [3] J. Lundell, F. Verdoja, and N. Le, "Constrained Generative Sampling of 6-DoF Grasps," 2023.
- [4] C. Eppner, A. Mousavian, and D. Fox, "ACRONYM: A Large-Scale Grasp Dataset Based on Simulation," 2020. [Online]. Available: <https://arxiv.org/pdf/2011.09584>